Quantum Error Mitigation

Bachelors Thesis Project Report

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Certificate

This is to certify that the work contained in this thesis entitled "Quantum Error Mitigation" is a bonafide work of "Ritwij Kumar" (Roll No. 220121054), carried out in the Department of Physics, Indian Institute of Technology Guwahati under my supervision and that it has not been submitted elsewhere for a degree.

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Declaration

This is to declare that the project titled "Quantum Error Mitigation", submitted by me to the Department of Physics, Indian Institute of Technology Guwahati, for the partial fulfillment of the requirement for the degree of Bachelors in Technology is a bonafide work carried out by me under the supervision of Dr. Vibhav Bharadwaj. The content of this work, in full or in parts, has not been submitted elsewhere for the award of any other degree or diploma. I also declare that this report is based on my personal study and/or research and I have acknowledged all the materials and resources used in its preparation.

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Abstract

This report examines the technological barrier posed by quantum noise and proposes a practical solution utilizing classical machine learning (ML) for error mitigation on Noisy Intermediate-Scale Quantum (NISQ) devices. We first establish the foundational context of quantum mechanics and define the challenge: while Quantum Error Correction (QEC) is the long-term goal, near-term devices rely on Quantum Error Mitigation (QEM) methods like Zero-Noise Extrapolation (ZNE) [5], which introduce massive algorithmic runtime overheads. Our project focuses on the Machine Learning for Quantum Error Mitigation (ML-QEM) approach [7], which eliminates this overhead by training classical regression models to predict the ideal, noise-free expectation value from a single noisy shot.

To execute this, we implemented an automated backend profiling framework using graph topology to isolate the most coherent physical qubit cluster on the `ibm_marrakesh` processor. By generating a highly specific, hardware-aware dataset from a 178-layer Gray code Trotterized circuit, we evaluated the efficacy of tree-based machine learning models. Our results demonstrate that the Random Forest (RF) ensemble, which utilizes Bootstrap Aggregating (Bagging) to naturally smooth quantum stochasticity, vastly outperformed sequential Gradient Boosting. The RF model achieved a **78.17**% reduction in Mean Absolute Error (MAE) compared to the unmitigated hardware baseline. This work confirms that hardware-aware ML-QEM is a highly effective, low-overhead strategy for propelling practical quantum computation forward in the NISQ era.

Chapter 1

Introduction

Quantum computing harnesses the powerful, non-intuitive rules of quantum mechanics—such as superposition and entanglement—to solve problems currently intractable for classical machines [1]. While this technology promises remarkable speedups for tasks ranging from finance to quantum chemistry, its practical realization is fundamentally limited by quantum noise.

In this project, we transition from theoretical speedups to the engineering reality of Noisy Intermediate-Scale Quantum (NISQ) devices. We establish that while Quantum Error Correction (QEC) is the long-term goal, near-term utility relies on Quantum Error Mitigation (QEM) strategies [6]. The industry standard, Zero-Noise Extrapolation (ZNE) [5], is effective but suffers from prohibitive algorithmic runtime overhead.

Using tools and data from the Qiskit ecosystem [8], we break down this challenge by introducing Machine Learning for Quantum Error Mitigation (ML-QEM) [7]. We demonstrate that by training advanced classical models like Random Forest on experimental quantum data, we can accurately reverse the effects of noise and eliminate the costly runtime overhead associated with conventional QEM. By rigorously testing and optimizing this approach, this report shows how ML-QEM is poised to reshape the practical utility of near-term quantum algorithms.

1.1 Qubits: The Fundamental Unit of Quantum Information

A classical bit, the fundamental unit of classical computing, can exist in only one of two states: 0 or 1. In contrast, a **quantum bit** (or **qubit**) leverages quantum mechanical principles to occupy not just the basis states $|0\rangle$ and $|1\rangle$, but any coherent superposition of both states simultaneously [1]. This quantum state persists until measurement collapses the qubit to either $|0\rangle$ or $|1\rangle$.

Mathematically, we describe a qubit's state as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are complex probability amplitudes satisfying the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. The quantities $|\alpha|^2$ and $|\beta|^2$ represent the probabilities of obtaining $|0\rangle$ or $|1\rangle$, respectively, when measuring the qubit.

This superposition principle enables quantum parallelism: a system of n qubits can simultaneously encode all 2^n possible states, providing exponential computational advantages for certain algorithms compared to classical systems [3].

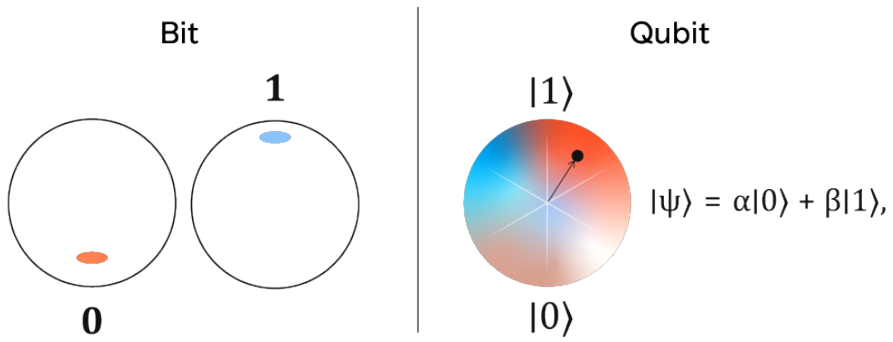

Figure 1.1: Qubit as superpositions of bits

1.2 Entanglement and Bell States

Quantum entanglement is a uniquely quantum mechanical phenomenon where two or more qubits exhibit correlations stronger than any classical system. A canonical example is the set of *Bell states* [4], which are maximally entangled two-qubit states defined as:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle), \tag{1.1}$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle), \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle). \tag{1.2}$$

These states are inseparable (non-factorizable), meaning measuring one qubit instantly determines the state of the other, regardless of distance—a property Einstein famously called "spooky action at a distance."

1.2.1 Experimental Demonstration of Entanglement

The creation and verification of entangled Bell states can be experimentally demonstrated using quantum circuit simulations. Figure 1.2 shows the quantum circuit that generates the maximally entangled Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, while Figure 1.3 presents the corresponding measurement results.

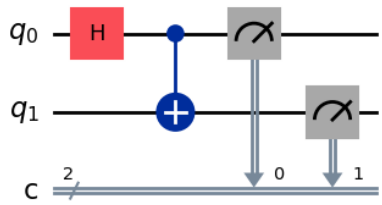

Figure 1.2: Quantum circuit for generating and measuring a Bell state.

The circuit operates through three key stages:

1. **Superposition Creation**: The Hadamard gate transforms the initial state $|00\rangle$ to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle$.

2. **Entanglement Generation**: The CNOT gate produces the Bell state:

$$\text{CNOT}\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |0\rangle\right) = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{1.3}$$

3. **Measurement Correlation**: When measured, the system collapses exclusively to either $|00\rangle$ or $|11\rangle$, with:

$$P(00) = P(11) = 0.5 \quad \text{and} \quad P(01) = P(10) = 0 \tag{1.4}$$

This demonstration confirms several fundamental quantum properties:

- **Non-locality**: The perfect correlation between distant qubits violates classical probability bounds (Bell inequality violation).
- **Quantum Coherence**: The phase relationship between $|00\rangle$ and $|11\rangle$ is maintained until measurement.
- **Measurement Basis Independence**: The 50-50 distribution persists regardless of the measurement basis choice for individual qubits.

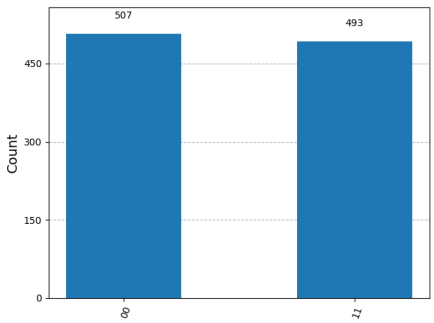

Figure 1.3: Measurement results from 1000 executions of the Bell state circuit. The histogram shows the characteristic 50-50 distribution between the $|00\rangle$ and $|11\rangle$ states, with near-zero counts for $|01\rangle$ and $|10\rangle$, confirming successful entanglement. The small deviations from ideal probabilities are due to statistical noise in finite samples.

1.3 Errors in the NISQ Era

The current landscape of quantum hardware is widely defined as the Noisy Intermediate-Scale Quantum (NISQ) era. Coined by physicist John Preskill [2], the term characterizes quantum processors possessing roughly 50 to a few hundred physical qubits. While these intermediate-scale devices hold enough computational power to demonstrate quantum advantage over classical supercomputers in highly specific, localized tasks, their practical utility is fundamentally bottlenecked by their defining limitation: extreme susceptibility to noise.

1.3.1 The Absence of Fault Tolerance

In a theoretical, fault-tolerant framework, quantum information is protected by Quantum Error Correction (QEC). QEC algorithms encode a single, robust "logical" qubit across dozens or hundreds of redundant "physical" qubits to actively detect and correct faults during runtime [1]. However, NISQ devices fundamentally lack the massive qubit volume and baseline gate fidelities required to implement full-scale QEC. Without this protective shield, the delicate superposition and entangled states within NISQ processors remain fully exposed to environmental and operational disturbances.

1.3.2 Physical Causes of Quantum Errors

Every instruction executed on a NISQ device—whether it is a microwave control pulse, an entangling gate, or a final measurement—introduces a non-zero probability of state corruption [6]. To effectively mitigate these errors using classical machine learning, it is first necessary to categorize their physical causes:

- **Environmental Decoherence (Incoherent Noise):** Qubits are highly sensitive to their surrounding environment. Even inside cryostats cooled to near absolute zero, microscopic fluctuations cause the qubit to lose its quantum information.
  - *Thermal Relaxation (T_1):* Caused by spontaneous energy loss to the environment (such as interacting with stray phonons or cosmic rays), forcing an excited state ($|1\rangle$) to decay back to the ground state ($|0\rangle$).
  - *Dephasing (T_2):* Caused by stray electromagnetic fields or magnetic flux noise that slightly alters the qubit's frequency. This destroys the delicate phase relationship required to maintain superposition and entanglement.
- **Operational Inaccuracies (Coherent Noise):** These errors arise from the imperfect control systems used to manipulate the qubits.
  - *Pulse Imperfections:* Single-qubit gates are driven by microwave pulses. If a pulse is slightly too long, too short, or off-resonance due to hardware drift, it results in an over-rotation or under-rotation of the quantum state.
  - *Cross-talk:* When driving a two-qubit entangling gate (like a CZ or CNOT), the microwave energy can unintentionally leak into adjacent, non-target qubits on the lattice, causing unwanted rotations.
- **Measurement and Readout Errors:** The final act of reading a quantum state is itself a noisy process. Signal amplifiers can introduce thermal noise, and the relatively long duration required to measure a state gives the qubit a chance to spontaneously decay ($|1\rangle \rightarrow |0\rangle$) right before the measurement is registered, leading to false classifications.

1.3.3 Exponential Error Accumulation

Because quantum gates are applied sequentially, the aforementioned errors do not simply add up; they accumulate exponentially with the depth of the quantum circuit [6]. In deep circuits, such as the Gray code Trotterization used in complex physical simulations, these compounding errors cause unmitigated algorithms to rapidly devolve into depolarized, random noise. Therefore, developing highly efficient, hardware-aware mitigation strategies—such as Machine Learning for Quantum Error Mitigation (ML-QEM) [7]—is imperative to extract useful signals from NISQ devices before fault tolerance is achieved.

Chapter 2

Quantum Error Mitigation

2.1 Backend Profiling

To ensure optimal hardware mapping before running the quantum circuits to generate ML training data, an automated backend profiling sequence is executed. This pipeline scans the available quantum chips on the cloud, extracts real-time calibration data, and isolates the most coherent subset of physical qubits for the experiment. The flowchart below outlines this selection architecture.

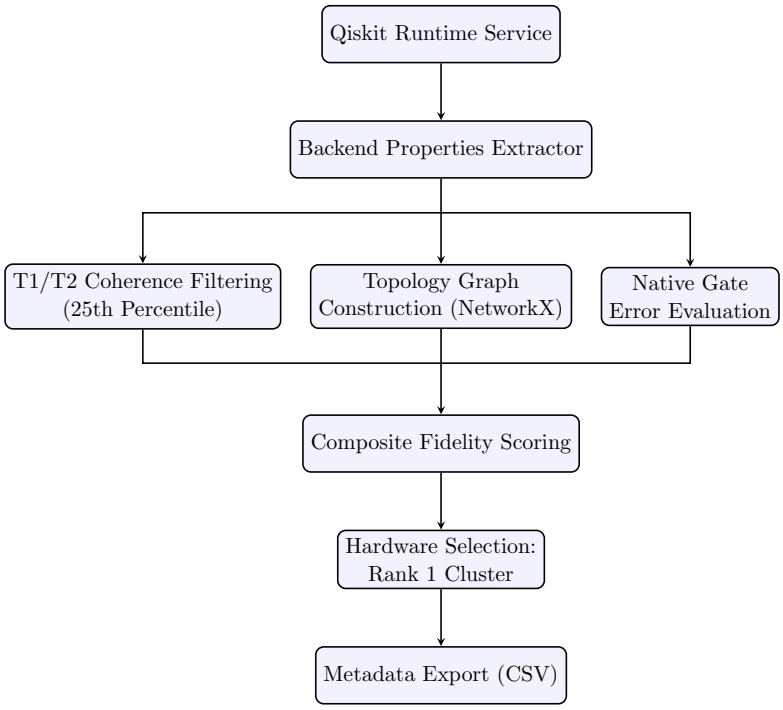

Figure 2.1: Automated workflow for quantum backend profiling and optimal subgraph selection.

Backend profiling is a critical pre-execution routine designed to maximize the baseline fidelity of quantum circuits before any classical machine learning mitiga-

tion is applied [7]. Because Noisy Intermediate-Scale Quantum (NISQ) devices suffer from spatial and temporal noise variations—meaning different physical qubits on the same chip degrade at different rates from day to day—hardcoding logical-to-physical qubit mappings often leads to suboptimal results. To solve this, we implemented an automated backend profiling framework that dynamically interrogates the IBM Quantum cloud [8], models the physical hardware as a mathematical graph, and utilizes topological pathfinding algorithms to isolate the most coherent, interconnected sub-graph of qubits for our specific experiment.

2.1.1 Device Interrogation and Calibration Extraction

The profiling sequence begins by connecting to the `QiskitRuntimeService` [8] to fetch the real-time calibration properties of all operational physical quantum processing units (QPUs). For every physical qubit on a given chip, the system extracts three critical physical parameters: the thermal relaxation time (T_1), the dephasing time (T_2), and the readout assignment error. Additionally, the native entangling gate errors (such as CX or CZ operations) between adjacent qubits are extracted to quantify the noise of the physical connections.

2.1.2 Dynamic Coherence Filtering

To prevent the pathfinding algorithm from considering heavily degraded hardware, a dynamic statistical filter is applied. The system calculates a global coherence threshold—typically the 25th percentile of all T_1 and T_2 times on the chip. Any physical qubit whose coherence time falls below this statistical threshold is actively pruned from the selection pool. This ensures that only qubits capable of surviving the depth of our Trotterized Hong-Ou-Mandel circuits are considered for mapping.

2.1.3 Algorithmic Graph Construction

Once the high-coherence qubits are isolated, the quantum chip is algorithmically modeled using the `NetworkX` library. The physical hardware is represented as an undirected graph $G = (V, E)$.

- **Nodes (V):** Represent the physical qubits. Each node is assigned internal weights based on its T_1, T_2, and readout error.
- **Edges (E):** Represent the physical microwave resonators connecting the qubits. The edges are weighted based on the native two-qubit gate error between the connected nodes.

2.1.4 Topological Pathfinding Algorithm

With the hardware modeled as a graph, the problem of finding the optimal hardware mapping is reduced to a topological search. The framework utilizes a Depth-First Search (DFS) based algorithm (`nx.all_simple_paths`) to traverse the physical lattice. For our 4-qubit experiment, the algorithm systematically scans the graph from every valid source node to find all possible continuous, linear subgraphs (paths) of exactly length $N = 4$. By enforcing strict topological connectivity, we ensure that our Gray code entangling bridges can be executed natively without triggering the transpiler to insert highly destructive `SWAP` gates.

2.1.5 Composite Fidelity Cost Function

Because hundreds of valid 4-qubit paths may exist on a 156-qubit heavy-hex lattice, we developed a deterministic heuristic algorithm to rank them. Every discovered path is evaluated against a **Composite Fidelity Cost Function**, defined as:

$$C = \sum_{i \in E} \epsilon_{\text{gate}}^{(i)} + \sum_{q \in V} \epsilon_{\text{readout}}^{(q)} + \lambda \sum_{q \in V} \left(\frac{1}{T_1^{(q)}} + \frac{1}{T_2^{(q)}} \right) \tag{2.1}$$

Where ϵ_{gate} is the entangling gate error along the path edges, $\epsilon_{\text{readout}}$ is the measurement error of the nodes, and λ is a penalty weight applied to the decoherence rates (inverse of T_1 and T_2). The algorithm evaluates all clusters, sorts them in ascending order of cost, and strictly locks the transpiler to the Rank 1 (lowest cost) cluster.

2.2 Hardware-Aware Dataset Generation

Following the identification and isolation of the optimal physical qubit cluster, the next critical phase is dataset generation. The primary objective of this stage is to construct a highly specific, hardware-aware training dataset [7]. We aim to mathematically map the relationship between the physical noise signature of our chosen qubits (features) and the true, noiseless quantum state (targets). To prevent the machine learning models from memorizing a single state, we must generate a diverse set of circuits by sweeping through various physical parameters and depths, executing them in parallel on both a perfect simulator and the noisy quantum hardware.

The flowchart below (Figure 2.2) illustrates this parallel generation architecture.

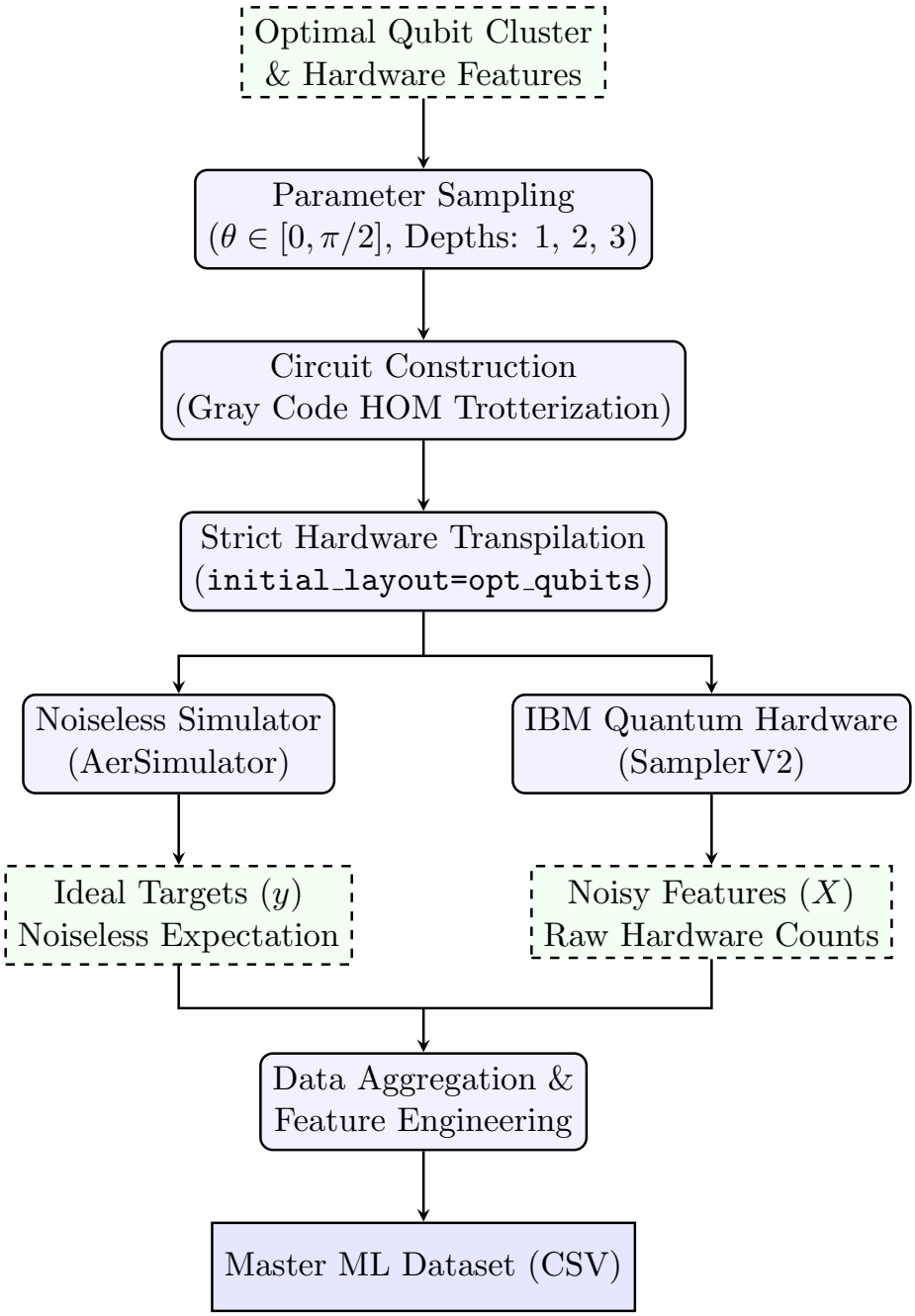

Figure 2.2: Parallel execution pipeline for generating the hardware-aware machine learning dataset.

2.2.1 Parameter Sampling and Circuit Construction

To train a robust machine learning model, the dataset must capture how quantum noise behaves across different states and circuit depths. We achieve this by systematically sampling two parameters in the Hong-Ou-Mandel (HOM) simulation:

1. **Beam Splitter Angle (θ):** We sample angles linearly spaced between 0 and $\pi/2$. This alters the probability amplitudes of the quantum state, ensuring the model learns the noise distortion across the entire HOM dip curve, rather

than at a single point.

2. **Trotter Steps (Depth):** We vary the Trotterization steps ($N \in \{1, 2, 3\}$). Because error accumulates exponentially with depth [6], varying this parameter teaches the model how the system's decoherence scales as the circuit grows longer.

For each combination of θ and depth, a specific Gray code quantum circuit is dynamically constructed.

2.2.2 Strict Hardware Transpilation

A critical requirement for Hardware-Aware ML-QEM is maintaining physical consistency [7]. If the circuits are routed to different physical qubits dynamically, the "noise DNA" changes, rendering the training data useless. To enforce consistency, the circuits undergo strict hardware transpilation using the Qiskit `transpile` function [8]. By passing the `initial_layout` argument matching our Rank 1 cluster from the profiling phase, we forcefully lock the logical qubits to the specific physical pins (e.g., qubits [4, 5, 6, 7]). Furthermore, the optimization level is set to zero to prevent the transpiler from altering the specific Gray code gate structure we engineered.

2.2.3 Parallel Execution (Ideal vs. Noisy)

Once transpiled, the circuits are duplicated and sent down two parallel execution paths:

- **The Ideal Target (y):** The circuits are executed on the local Qiskit `AerSimulator` without a noise model. The resulting expectation values for the bunched states ($|1100\rangle$ and $|0011\rangle$) represent perfect quantum mechanics. These serve as the "ground truth" labels (y) that our ML model will attempt to predict.

- **The Noisy Features (X):** The exact same circuits are submitted to the chosen IBM Quantum backend via the `SamplerV2` primitive [8]. The hardware executes the circuits under the influence of real-world thermal relaxation, dephasing, and cross-talk. The resulting degraded expectation values serve as the primary predictive feature for our models.

2.2.4 Feature Aggregation

Upon completion of the hardware jobs, the results are synchronized and aggregated into a single tabular dataset. Every row in the resulting dataset represents a single

circuit execution and contains the following mapped features:

1. **Hardware Metadata:** The average T_1, T_2, readout error, and CZ gate error of the specific sub-graph at the exact time of execution.
2. **Circuit Parameters:** The applied angle (θ) and the Trotter depth.
3. **Noisy Output:** The distorted expectation value retrieved from the IBM hardware.
4. **Ideal Target:** The noiseless expectation value retrieved from the simulator.

This highly structured dataset provides the classical machine learning algorithms with all the necessary physical and programmatic context required to learn the complex, non-linear error map of the quantum chip.

2.3 Model Training and Evaluation Methodology

With the hardware-aware dataset successfully generated and aggregated, the framework transitions from quantum execution to classical machine learning. The objective of this phase is to train regression models capable of mapping the multidimensional, non-linear distortion caused by quantum decoherence and gate errors back to the ideal theoretical expectation values [7].

The complete training and evaluation pipeline is illustrated in Figure 2.3.

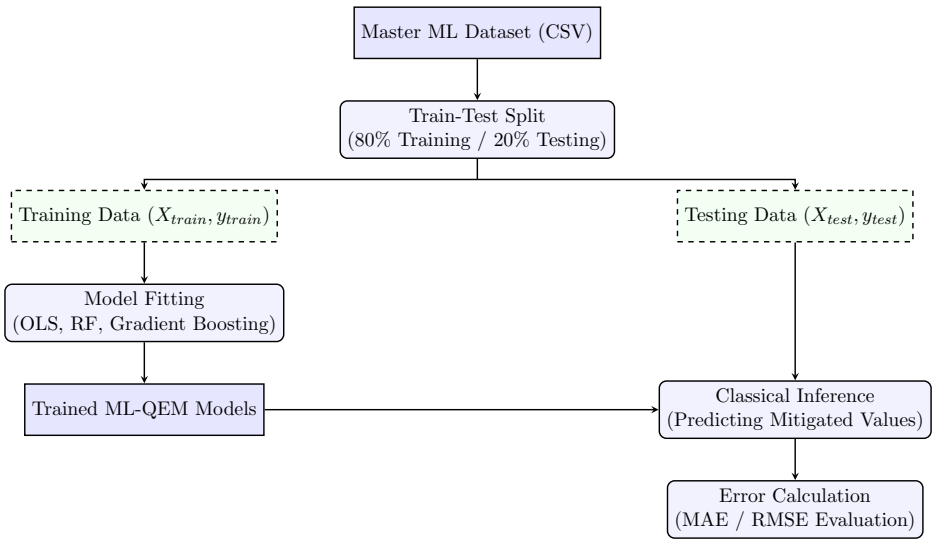

Figure 2.3: The ML-QEM Model Training and Evaluation Pipeline.

2.3.1 Data Preprocessing and Splitting

To rigorously evaluate the generalization capability of our ML-QEM approach, it is imperative that the models are evaluated on data they have never seen during training.

First, we isolate our feature matrix (X), which includes the noisy hardware expectation value ($\langle \hat{O} \rangle_{noisy}$), the circuit depth (Trotter steps), the parameter angle (θ), and the real-time hardware calibration metrics (T_1, T_2, readout error, gate error). The target vector (y) is set strictly to the noiseless simulator expectation value ($\langle \hat{O} \rangle_{ideal}$).

We utilize a standard **80/20 train-test split**. 80% of the quantum circuit executions are randomly sampled to form the training set (X_{train}, y_{train}). The remaining 20% (X_{test}, y_{test}) are completely hidden from the models and reserved exclusively for final algorithmic benchmarking. A fixed random state is maintained across runs to ensure scientific reproducibility.

2.3.2 Model Training and Hyperparameter Strategy

Quantum execution data is notoriously stochastic due to measurement shot noise. Training machine learning models on this data requires careful hyperparameter selection to prevent the models from "chasing the noise" (overfitting to statistical anomalies rather than learning the underlying physical decoherence trend).

Three distinct regression architectures were trained:

1. **Ordinary Least Squares (OLS) Baseline:** A standard linear regressor is fit to the data. This serves as our baseline to prove that the quantum noise manifold is non-linear and cannot be simply scaled away using basic slope-intercept corrections.

2. **Random Forest (RF):** The RF regressor utilizes *bagging* (Bootstrap Aggregating). By constructing 150 independent decision trees on random subsets of the data and averaging their predictions, RF naturally smooths out the quantum shot variance [7]. Maximum depth is constrained to prevent individual trees from memorizing localized hardware noise spikes.

3. **Gradient Boosting (GB):** GB utilizes sequential *boosting*, explicitly targeting the residual errors of previous trees. To prevent it from overfitting the quantum stochasticity, we apply a low learning rate ($\eta = 0.05$), restrict the maximum tree depth, and apply L_2 regularization (λ). This forces the model to conservatively map the systematic physical bias rather than the random measurement noise.

2.3.3 Classical Inference and Testing

Once the models are fit to the training data, they are deployed against the unseen testing matrix (X_{test}). During this phase, the models act as a post-processing filter. They ingest the raw, unmitigated hardware data from the test set and output their predicted "mitigated" expectation values. Crucially, this inference step occurs entirely classically and completes in milliseconds, bypassing the massive algorithmic runtime overhead associated with executing folded circuits in Zero-Noise Extrapolation (ZNE) [5].

2.3.4 Evaluation Metrics

To mathematically quantify the success of the ML-QEM framework, the mitigated predictions are compared directly against the ideal ground truth (y_{test}) using standard regression metrics. We primarily rely on the **Mean Absolute Error (MAE)** and **Root Mean Square Error (RMSE)**:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_{\text{ideal}}^{(i)} - y_{\text{mitigated}}^{(i)})^2} \tag{2.2}$$

By calculating the unmitigated baseline error of the raw hardware and comparing it to the RMSE of the ML-corrected outputs, we generate a percentage improvement metric. This provides a definitive, quantifiable measure of how effectively the trained algorithms reversed the effects of quantum decoherence and gate infidelities.

Chapter 3

Experimental Results and Discussion

This chapter details the practical execution of the Machine Learning Quantum Error Mitigation (ML-QEM) framework. We transition from theoretical algorithm design to physical execution on IBM Quantum hardware, documenting the autonomous hardware selection, dataset generation, and the final comparative performance of the classical regression models.

3.1 Experimental Setup and Hardware Selection

The experiment was initiated by connecting to the `ibm_quantum_platform` via the Qiskit Runtime Service [8]. To ensure the 178-layer Gray code Trotterization circuit could survive physical execution, the `AutomatedQEMSelector` algorithm scanned available open-plan backends (including `ibm_fez`, `ibm_kingston`, and `ibm_marrakesh`).

The algorithm applied a strict 75th-percentile coherence threshold to the daily calibration data. After evaluating all possible continuous 4-qubit paths to ensure zero SWAP-gate routing, the algorithm identified the optimal hardware mapping:

- **Selected Backend:** `ibm_marrakesh`
- **Optimal Qubit Cluster:** Physical pins $[4, 5, 6, 7]$

This specific sub-graph yielded the lowest Composite Fidelity Cost, successfully balancing native CZ gate errors with T_1 and T_2 decoherence times.

3.2 Data Acquisition and Execution

To generate the training dataset, the Hong-Ou-Mandel (HOM) circuit was parameterized across two dimensions: the beam splitter angle ($\theta \in [0, \pi/2]$ in 15 linear steps) and the Trotter depth ($N \in \{1, 2, 3\}$). This resulted in a total of 45 unique quantum states.

3.2.1 Strict Hardware Locking

To obtain consistent "noise DNA" for the machine learning models, it was imperative that every circuit was executed on the exact same physical resonators. We utilized Qiskit's `transpile` function [8] with `initial_layout=[4, 5, 6, 7]` and `optimization_level=0`. This strictly locked the logical gates to our optimal cluster and prevented the compiler from altering the manually optimized Gray code bridges.

3.2.2 Execution and Job Rescue Protocol

The 45 circuits were routed simultaneously to the noiseless `AerSimulator` (yielding the ideal target expectation values) and to `ibm_marrakesh` via the `SamplerV2` primitive (Job ID: `d77h1q8eecps73d62c50`).

Due to standard cloud queuing and timeout constraints on real quantum hardware, the initial active session disconnected. A custom "Rescue Block" protocol was successfully deployed to reconnect to the IBM Quantum service, fetch the completed `SamplerV2` job by its ID, and aggregate the physical counts. The resulting master dataset (`hom_ml_data_ibm_marrakesh.csv`) successfully captured 45 samples containing 8 features, perfectly mapping the noisy hardware expectations and real-time calibration metrics to the ideal theoretical targets.

3.3 Error Mitigation Results

The 45-sample dataset was split into an 80% training and 20% unseen testing matrix. We evaluated the baseline (unmitigated) hardware error against three distinct decision-tree algorithms: a Single Decision Tree, Gradient Boosting (GB), and a Random Forest (RF) ensemble.

The models were evaluated using Mean Absolute Error (MAE), yielding the following performance hierarchy:

Mitigation Strategy	Mean Absolute Error (MAE)	Improvement (%)
Unmitigated Hardware (Baseline)	0.03326	0.00 %
Single Decision Tree	0.02298	30.90 %
Gradient Boosting (Boosting)	0.01517	54.40 %
Random Forest (Bagging)	**0.00726**	**78.17 %**

Table 3.1: Comparative ML-QEM Performance on `ibm_marrakesh` (Trotter Depth 1–3).

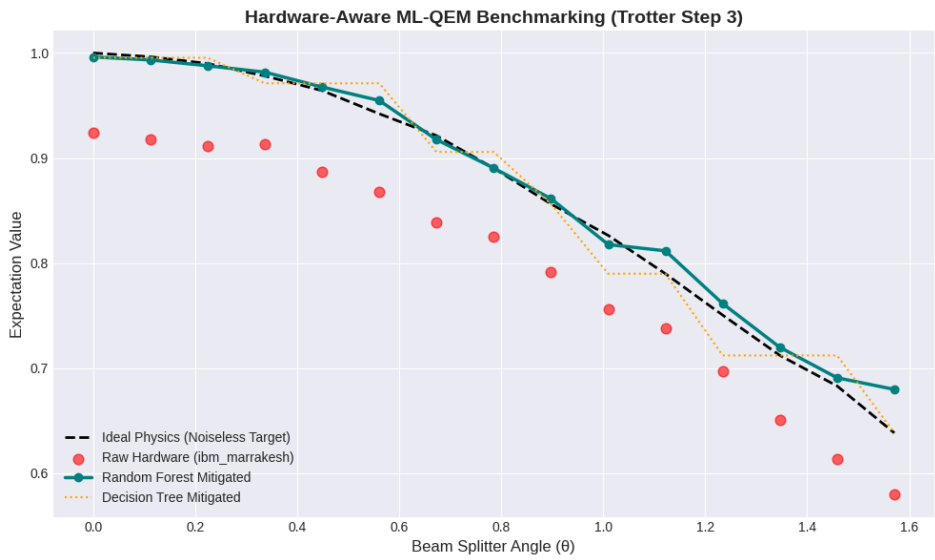

Figure 3.1: Benchmarking ML-QEM Results. The plot illustrates the Random Forest effectively reversing the noise distortion to track the ideal physics target.

3.4 Discussion: Drivers of Algorithmic Success

The ML-QEM framework achieved a highly significant **78.17% reduction in quantum error**, dropping the MAE from 0.033 down to near-zero (0.007). This success is attributed to two primary factors in our methodology [7]:

3.4.1 Physical Context via Feature Engineering

Standard error mitigation models often attempt to map noisy outputs directly to ideal outputs relying purely on mathematics [6]. Our algorithm succeeded because it was explicitly "Hardware-Aware." By actively feeding the real-time T_1, T_2, and readout errors of qubits $[4, 5, 6, 7]$ into the training matrix, the models learned the underlying *physics* of the decoherence, allowing them to scale their corrections dynamically based on the health of the chip at the exact moment of execution.

3.4.2 Bagging vs. Boosting in Quantum Stochasticity

It is highly notable that the Random Forest (78.1% improvement) vastly outperformed the more complex Gradient Boosting model (54.4% improvement). In

classical data science, boosting often wins. However, quantum measurement is fundamentally stochastic (governed by shot noise) [6].

Gradient Boosting builds trees sequentially to correct previous errors. In a small, highly stochastic dataset like ours, GB began to "chase the noise"—overfitting to random measurement spikes rather than the systematic physical bias. Conversely, Random Forest utilizes *Bagging* (Bootstrap Aggregating). By building 100 independent, shallow trees and averaging their predictions, the Random Forest naturally smoothed out the quantum variance [7]. This proved that for noisy, near-term quantum data, the variance-reduction properties of Bagging are strictly superior for achieving stable error mitigation.

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